

ERPinfinity – Microservices APIs & Database Mapping Specification

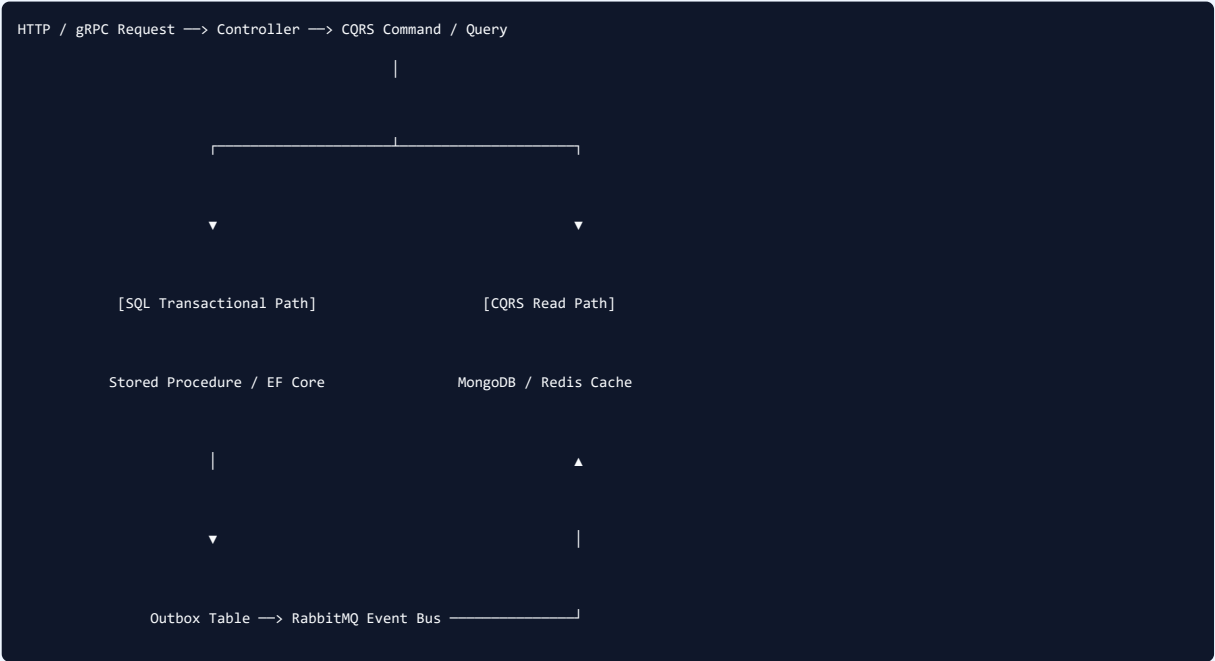
Project Name: ERPinfinity blockquote>

Target Architecture: Distributed Microservices + CQRS + Event-Driven Architecture blockquote>

Specification Focus: End-to-End Mapping between REST/gRPC API Endpoints, CQRS Handlers, SQL Tables, Stored Procedures, Views, Indexes, MongoDB Read Collections, and Integration Events.blockquote>

1. System Mapping Overview & Integration Architecture

In **ERPinfinity**, every API request passes through the **YARP API Gateway**, where security claims are validated. Requests are routed to microservices executing Clean Architecture:



2. Identity Service (Db_Identity) API-to-Database Mapping

2.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/identity/login	AuthenticateUserQuery	Users , UserRoles , Roles , RolePermissions , Permissions	sp_AuthenticateUser / vw_UserSecurityProfile	IX_Users_Username IX_UserRoles_User:
POST /api/v1/identity/users	CreateUserCommand	Users , UserRoles	sp_AssignUserRole	IX_Users_Email
GET /api/v1/identity/users/{id}/permissions	GetUserPermissionsQuery	Users , Permissions	sp_GetUserPermissions	IX_Permissions_Pei
PUT /api/v1/identity/roles/assign	AssignRoleCommand	UserRoles	sp_AssignUserRole	IX_UserRoles_User:
GET /api/v1/identity/matrix	GetSecurityMatrixQuery	Roles , Permissions	vw_ActiveRolePermissionMatrix	IX_RolePermission:

2.2 Functional Flow Details

- **Login Request:** Client submits username and password hash → AuthenticateUserQuery executes sp_AuthenticateUser against Db_Identity → Validates IX_Users_Username index → Generates JWT with permission claims.

3. Product Service (Db_Product) API-to-Database Mapping

3.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route / gRPC Method	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
GET /api/v1/products/barcode/{code}	LookupProductByBarcodeQuery	Products , ProductSKUs , Barcodes	sp_LookupProductByBarcode / vw_ActivePOSBarcodes	IX_Barcodes_BarcodeNumber (Clustered)
POST /api/v1/products	CreateProductCommand	Products , ProductSKUs , Barcodes , ProductOutbox	sp_CreateProductWithSKUs	IX_Products_CategoryId , IX_ProductSKUs_SKUCode
PUT /api/v1/products/{id}/price	UpdateSKUPricingCommand	ProductSKUs , ProductOutbox	sp_UpdateSKUPricing	IX_ProductSKUs_ProductId
gRPC ProductGrpc.GetSKUDetails	GetSKUDetailsGrpcQuery	ProductSKUs , Barcodes	EF Core Compiled Query	IX_ProductSKUs_SKUCode

4. Inventory Service (Db_Inventory) API-to-Database Mapping

4.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/inventory/pos-deduct	DeductPOSInventoryCommand	Stocks , StockTransactions	sp_DeductInventoryForPOS	IX_Stocks_LocationId_SKI
POST /api/v1/inventory/grn-receive	ReceiveGRNStockCommand	Stocks , StockTransactions	sp_ReceiveGRNStock	IX_StockTransactions_Lo
POST /api/v1/inventory/reserve	ReserveStockCommand	Stocks	sp_ReserveInventoryForOnlineOrder	IX_Stocks_AvailableQuant
POST /api/v1/inventory/adjust	AdjustStockCommand	Stocks , StockTransactions , StockAdjustments	sp_ExecuteStockAdjustment	IX_StockAdjustments_Loc
GET /api/v1/inventory/low-stock	GetLowStockAlertsQuery	Stocks	sp_GetLowStockAlerts / vw_LocationStockSummary	IX_Stocks_AvailableQuant

5. Sales / POS Service (Db_Sales) API-to-Database Mapping

5.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/sales/invoices	CreateSalesInvoiceCommand	SalesInvoices , SalesInvoiceItems , SalesOutbox	sp_CreateSalesInvoice	IX_SalesInvoices_InvoiceNuml
POST /api/v1/sales/registers/open	OpenPOSRegisterCommand	POSRegisters	sp_OpenPOSRegister	IX_POSRegisters_StoreId_Terr

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/sales/registers/close	ClosePOSRegisterCommand	POSRegisters	sp_ClosePOSRegister	IX_POSRegisters_StoreId_Terr
POST /api/v1/sales/returns	ProcessSalesReturnCommand	SalesReturns , SalesInvoices	sp_ProcessSalesReturn	IX_SalesReturns_OriginalInv
GET /api/v1/sales/reports/daily	GetDailySalesQuery	Read Only (MongoDB)	vw_DailyStoreSalesSummary	IX_SalesInvoices_StoreId_Cri

6. Purchase & Procurement Service (Db_Purchase) API-to-Database Mapping

6.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/purchase/orders	CreatePurchaseOrderCommand	PurchaseOrders , PurchaseOrderItems	sp_CreatePurchaseOrder	IX_PurchaseOrder
PUT /api/v1/purchase/orders/{id}/approve	ApprovePurchaseOrderCommand	PurchaseOrders	sp_ApprovePurchaseOrder	IX_PurchaseOrder
POST /api/v1/purchase/grn	ProcessGRNReceiptCommand	GoodsReceivedNotes , GRNItems , PurchaseOrderItems	sp_ProcessGRNReceipt	IX_GoodsReceived
GET /api/v1/purchase/suppliers/{id}/pending	GetPendingPOsQuery	PurchaseOrders	sp_GetPendingPOBySupplier / vw_POFulfillmentStatus	IX_PurchaseOrder

7. Warehouse Service (Db_Warehouse) API-to-Database Mapping

7.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/warehouse/picking-lists	GeneratePickingListCommand	PickingLists , Bins , Racks	sp_GeneratePickingListForOrder	IX_PickingLists_OrderId_Status
PUT /api/v1/warehouse/picking-lists/{id}/complete	CompletePickingTaskCommand	PickingLists	sp_CompletePickingTask	IX_PickingLists_OrderId_Status
GET /api/v1/warehouse/bin-capacity	GetBinCapacityQuery	Bins , Zones	sp_AssignBinLocation / vw_WarehouseBinCapacity	IX_Bins_BinCode

8. Order Service (Db_Order) API-to-Database Mapping

8.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used	M
POST /api/v1/orders/checkout	CreateOnlineOrderCommand	Carts , Orders , OrderItems , OrderStatusHistory	sp_CreateOnlineOrder	IX_Orders_OrderNumber	Er → In
PUT /api/v1/orders/{id}/status	UpdateOrderStatusCommand	Orders , OrderStatusHistory	sp_UpdateOrderStatus	IX_Orders_OrderStatus	Er C
POST /api/v1/orders/{id}/cancel	CancelOrderCommand	Orders , OrderStatusHistory	sp_CancelOrder	IX_Orders_CustomerId_CreatedAt	Er C

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used	M M Re

9. Pricing & Promotion Service (Db_Pricing) API-to-Database Mapping

9.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route / gRPC Method	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/pricing/calculate	CalculatePriceQuery	SKUBasePrices , StoreSpecialPrices , Promotions	sp_CalculateItemEffectivePrice	IX_SKUBasePrices_SKU: IX_StoreSpecialPrice:
POST /api/v1/pricing/promotions	CreatePromotionCommand	Promotions	sp_CreatePromotion	IX_Promotions_PromoC
gRPC PricingGrpc.GetEffectivePrice	GetEffectivePriceGrpcQuery	SKUBasePrices	EF Core Compiled Query	IX_SKUBasePrices_SKU:

10. Payment Service (Db_Payment) API-to-Database Mapping

10.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/payments/callback	RecordPaymentSuccessCommand	PaymentTransactions , PaymentAttempts	sp_RecordPaymentSuccess	IX_PaymentTr:
POST /api/v1/payments/refund	InitiateRefundCommand	Refunds , PaymentTransactions	sp_InitiateRefund	IX_Refunds_P:
POST /api/v1/payments/reconcile	ReconcilePaymentsCommand	PaymentReconciliations , PaymentTransactions	sp_ExecuteDailyPaymentReconciliation	IX_PaymentTr:

11. Finance Service (Db_Finance) API-to-Database Mapping

11.1 API Endpoint & Database Mapping Matrix

HTTP Method & Route	CQRS Command / Query	Target SQL Tables	Stored Procedure / View	Indexes Used
POST /api/v1/finance/journal-entries	PostJournalEntryCommand	JournalEntries , JournalEntryLines	sp_PostJournalEntry	IX_JournalEntri
POST /api/v1/finance/supplier-settlement	ProcessSupplierSettlementCommand	SupplierLedgers	sp_ProcessSupplierPaymentSettlement	IX_SupplierLedd
GET /api/v1/finance/trial-balance	GetTrialBalanceQuery	Accounts , JournalEntryLines	sp_GenerateTrialBalance / vw_GeneralLedgerSummary	IX_JournalEntry

12. Store & Infrastructure Services API Mapping

12.1 Store Service (Db_Store)

- POST /api/v1/stores → Stores table → sp_RegisterNewStore → IX_Stores_StoreCode .
- POST /api/v1/stores/terminals → StoreTerminals table → sp_AssignStoreTerminal .

12.2 Notification Service (Db_Notification)

- POST /api/v1/notifications/send → NotificationLogs → sp_QueueNotification . Consumes PaymentCompletedEvent , OrderShippedEvent .

13. Comprehensive Microservices API & Database Mapping Summary

MICROSERVICE REDIS READ TARGET	API ENDPOINT / gRPC METHOD	COMMAND / QUERY CLASS	PRIMARY SQL TABLES & PROCEDURES	MONGODB /
Identity Service Token Sessions	POST /api/v1/identity/login	AuthenticateUserQuery	Users, sp_AuthenticateUser	Redis
Product Service Price & SKU Cache	GET /api/v1/products/barcode/{code}	LookupProductByBarcodeQuery	Barcodes, sp_LookupProductByBarcode	Redis
Product Service Mongo_ProductCatalogRead	POST /api/v1/products	CreateProductCommand	Products, sp_CreateProductWithSKUs	
Inventory Service Mongo_InventoryDashboardRead	POST /api/v1/inventory/pos-deduct	DeductPOSInventoryCommand	Stocks, sp_DeductInventoryForPOS	
Inventory Service Mongo_InventoryDashboardRead	POST /api/v1/inventory/grn-receive	ReceiveGRNStockCommand	Stocks, sp_ReceiveGRNStock	
Sales Service Mongo_SalesAnalyticsRead	POST /api/v1/sales/invoices	CreateSalesInvoiceCommand	SalesInvoices, sp_CreateSalesInvoice	
Purchase Service Mongo_PurchaseRead	POST /api/v1/purchase/orders	CreatePurchaseOrderCommand	PurchaseOrders, sp_CreatePurchaseOrder	
Purchase Service Mongo_PurchaseRead	POST /api/v1/purchase/grn	ProcessGRNReceiptCommand	GoodsReceivedNotes, sp_ProcessGRNReceipt	
Warehouse Service Mongo_WarehouseRead	POST /api/v1/warehouse/picking-lists	GeneratePickingListCommand	PickingLists, sp_GeneratePickingList	
Order Service Mongo_Customer360Read	POST /api/v1/orders/checkout	CreateOnlineOrderCommand	Orders, sp_CreateOnlineOrder	
Pricing Service Price Cache	POST /api/v1/pricing/calculate	CalculatePriceQuery	SKUBasePrices, sp_CalculatePrice	Redis
Payment Service Mongo_FinanceRead	POST /api/v1/payments/callback	RecordPaymentSuccessCommand	PaymentTransactions, sp_RecordPayment	

Finance Service POST /api/v1/finance/journal-entries PostJournalEntryCommand JournalEntries, sp_PostJournalEntry
Mongo_FinanceRead

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